Bahaa Samir

AI/ML Systems Engineer | Distributed AI Infrastructure

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OPEN-SOURCE ML SYSTEMS & INFRASTRUCTURE PROJECTS

Llama_Pure_Compute | High-Performance LLM Runtime & CUDA Kernel Library

- Developed fused C++20/CUDA transformer kernels achieving up to 3.56× faster RMSNorm/SwiGLU execution, 1.70× faster FlashAttention-v2, and 978.7 GB/s memory bandwidth through hardware-aware optimization.

VortexGrid-Engine | Distributed LLM Training Platform

- Architected a distributed LLM training platform supporting DDP, FSDP, DeepSpeed ZeRO, Tensor Parallelism, Ray orchestration, and fault-tolerant checkpointing, reducing GPU memory consumption by over 65% while enabling scalable multi-GPU and multi-node training.

NexusCache-Engine | Low-Latency C++/Python KV-Cache & Vector Storage System

- Engineered a LLM serving platform featuring dynamic paged KV-cache allocation, continuous batching, asynchronous Ray scheduling, and hardware-aware GPU memory management, achieving 92% GPU utilization, <35 ms p95 latency, and 2.4× higher serving throughput.

AetherScale-RAG | Distributed LLM Serving & Memory Infrastructure

- Designed a distributed retrieval platform combining hybrid search, native C++20 acceleration, parallel ingestion, and production observability, indexing 1M documents in 8.4 minutes while sustaining <14.2 ms p99 retrieval latency

EvalAgent-Platform | Multi-Agent Evaluation & RLHF Infrastructure

- Designed a production-scale multi-agent runtime separating reasoning, execution, memory, retrieval, and evaluation into independently scalable services, delivering 72% faster distributed evaluation, <15 ms streaming latency, and 10,000+ trajectory events/sec.

OmniPEFT | Distributed Fine-Tuning & Post-Training Platform

- Built a modular fine-tuning framework integrating QLoRA, CUDA optimization, statistical evaluation, experiment tracking, and production deployment to accelerate efficient post-training of large language models.

CORE TECHNICAL SKILLS

- **Languages:** C++20, Python, CUDA C/C++, Triton
- **Distributed AI:** PyTorch, DeepSpeed, FSDP, Megatron-LM, Ray, NVIDIA NCCL, Tensor Parallelism, Data Parallelism, Pipeline Parallelism,
- **LLM Systems:** vLLM, TensorRT-LLM, PagedAttention, Speculative Decoding, FP8/INT4 Quantization
- **Performance Engineering:** CUDA Streams, FlashAttention, RMSNorm, S w i G L U , Shared Memory, cuBLAS, Nsight
- **AI Infrastructure:** FastAPI, Apache Kafka, Redis, PostgreSQL, Qdrant, Docker, Kubernetes
- **MLOps & Cloud:** Git, CI/CD, MLflow, Weights & Biases, Prometheus, Grafana, Linux, AWS

PROFESSIONAL EXPERIENCE

AI Research & ML Systems Intern | Zagazig University | Jan 2026 – Aug 2026

- Built scalable ML inference and evaluation pipelines for large-scale genomic datasets, leveraging PyTorch, CUDA, multi-threading, and performance profiling to optimize throughput and model evaluation.
- Designed automated benchmarking workflows for model calibration, uncertainty estimation, and statistical validation across heterogeneous clinical datasets.

AI Research Intern – Computational Genomics | Ain Shams University (Demerdash) | Oct 2024 – Jan 2026

- Developed predictive machine learning models for genomic risk analysis using high-dimensional clinical and genetic datasets.
- Implemented high-performance C++/Python data pipelines for feature engineering, model evaluation, and large-scale genomic data processing.

EDUCATION

Faculty of Medicine, MTI University | Cairo, Egypt

Bachelor of Medicine and Surgery (M.B.B.S.) | 2019–2024

- Concurrent Academic Track in Artificial Intelligence & Computer Science (2021–2024), completing bachelor-level coursework in Machine Learning, Deep Learning, Data Structures & Algorithms, Systems Programming (C/C++), Operating Systems, Computer Architecture, Parallel & Distributed Computing, and Applied Mathematics.
- Computational specialization in Quantitative Genetics, Clinical Biostatistics, and High-Dimensional Disease Modeling.